



**KEMENTERIAN
PENDIDIKAN
MALAYSIA**

KOLEJ MATRIKULASI PERAK

PRA-PSPM
SET 1

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Name: _____

Matric. No.: _____

Class: _____

NO.	MARKS	
1	16	
2	12	
3	10	
4	14	
5	14	
6	14	
TOTAL	80	

INSTRUCTION TO CANDIDATE:

This question paper consists of **6** questions. Answer **all** the questions.
The use of electronic calculator is permitted.

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Elementary charge magnitude	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Rydberg constant	R_H	$= 1.097 \times 10^7 \text{ m}^{-1}$
Avogadro constant	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Gravitational constant	G	$= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Free-fall acceleration	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|---|---|
| 1. $x = A \sin \omega t$ | 21. $\phi = BA \cos \theta$ |
| 2. $\omega = \frac{2\pi}{T} = 2\pi f$ | 22. $\Phi = N\phi$ |
| 3. $k = \frac{2\pi}{\lambda}$ | 23. $\varepsilon = -\frac{d\phi}{dt}$ |
| 4. $v = f\lambda$ | 24. $\varepsilon = Blv \sin \theta$ |
| 5. $y(x, t) = A \sin(\omega t \pm kx)$ | 25. $\varepsilon = -NA \frac{dB}{dt}$ |
| 6. $C = \frac{Q}{V}$ | 26. $\varepsilon = -NB \frac{dA}{dt}$ |
| 7. $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$ | 27. $R = 2f$ |
| 8. $C = C_1 + C_2 + C_3 + \dots + C_n$ | 28. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ |
| 9. $\tau = RC$ | 29. $m = \frac{h_i}{h_o} = -\frac{v}{u}$ |
| 10. $Q = Q_o e^{-\frac{t}{RC}}$ | 30. $y_m = \frac{m\lambda D}{d}$ |
| 11. $Q = Q_o \left(1 - e^{-\frac{t}{RC}}\right)$ | 31. $y_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$ |
| 12. $I = \frac{dQ}{dt}$ | 32. $\Delta y = \frac{\lambda D}{d}$ |
| 13. $Q = ne$ | |
| 14. $V = IR$ | |
| 15. $\rho = \frac{RA}{l}$ | |
| 16. $R = R_1 + R_2 + R_3 + \dots + R_n$ | |
| 17. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ | |
| 18. $B = \frac{\mu_o I}{2\pi r}$ | |
| 19. $B = \frac{\mu_o I}{2r}$ | |
| 20. $B = \mu_o nI$ | |

1. (a) A sinusoidal wave is represented by the equation

$$y = 50 \sin \pi \left(\frac{1}{2} t - 2x \right)$$

where y and x are in cm and t is in seconds. Calculate the

- (i) period
- (ii) wavelength
- (iii) maximum vibrational velocity of the particle
- (iv) wave speed.

[8 marks]

- (b) The equation of a stationary wave along a stretched string is given as

$$y = 100 \cos \frac{\pi}{6} x \sin \pi t$$

where y and x are in meters and t is in seconds. Calculate the

- (i) frequency and wavelength of the wave
- (ii) speed of the progressive waves.

[4 marks]

- (c) A string of 1.5 m long and a mass of 100 grams has one of its ends fixed to a wall and the other end is forced to vibrate transversely with a frequency of 50 Hz. If the tension in the string is 300 N, calculate

- (i) mass per unit length
- (ii) velocity of the wave
- (iii) wavelength of the wave produced.

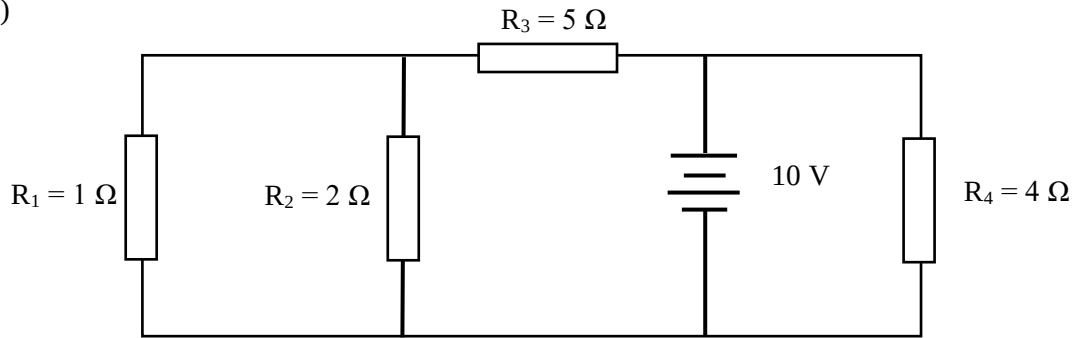
[4 marks]

2. (a) A wire 5 m long and has a radius of 0.2 mm with a resistance of 20Ω . A potential difference of 20 V is applied between the ends. Calculate the

- (i) current in the wire
- (ii) resistivity of the wire material.

[3 marks]

(b)

**FIGURE 1**

Four resistors are connected to a battery as shown in **FIGURE 1**. The resistance of R_1 is $1\ \Omega$, R_2 is $2\ \Omega$, R_3 is $5\ \Omega$ while R_4 is $4\ \Omega$. The electromotive force (e.m.f) of the battery is 10 V. Calculate the

- (i) effective resistance of the circuit, R_{eff}
- (ii) total current in the circuit, I_{total}
- (iii) voltage across resistor R_1 , V_1

[9 marks]

3. (a)

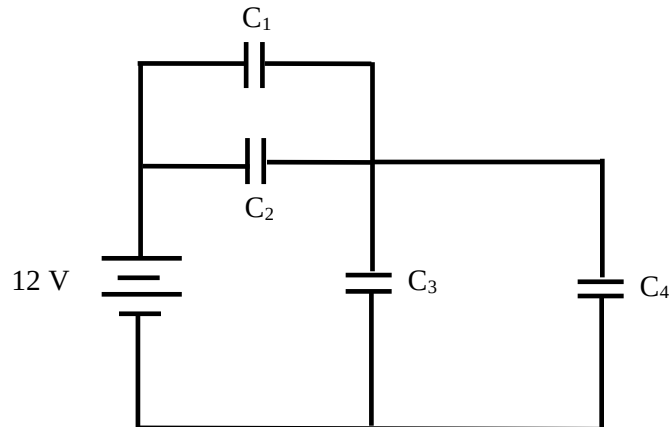
**FIGURE 2**

FIGURE 2 shows four identical capacitors of $6\ \mu\text{F}$ are connected in the circuit. Calculate the

- (i) effective capacitance
- (ii) charge on capacitor C_3

[4 marks]

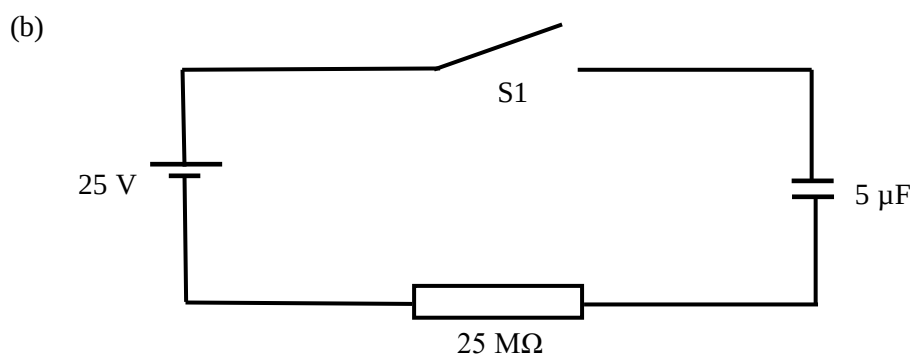
**FIGURE 3**

FIGURE 3 shows a direct current circuit. After the switch S_1 is closed,

- (i) calculate the time constant
- (ii) calculate the time taken for the charge to reach 25% of its maximum value
- (iii) sketch a graph of current against time.

[6 marks]

4. (a)

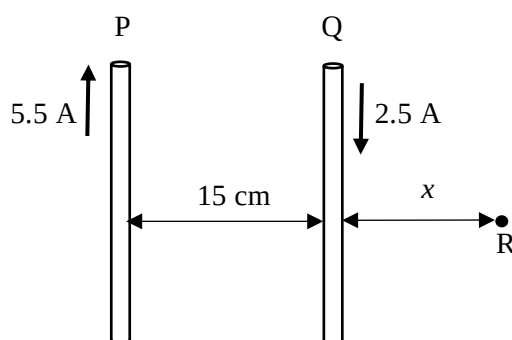
**FIGURE 4**

FIGURE 4 shows two parallel long wires P and Q which carries current, 5.5 A and 2.5 A respectively in the opposite directions. The wires are separated 15 cm apart. Calculate

- (i) value of x if the resultant magnetic field is zero at point R
- (ii) magnitude of the resultant magnetic field at midpoint between the wires.

[9 marks]

(b)

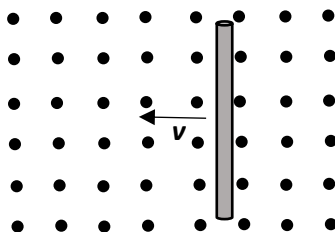


FIGURE 5

FIGURE 5 shows a 0.8 m long copper rod with resistance $25\ \Omega$ moves to the left with a speed of $7.5\ \text{m s}^{-1}$ perpendicular to a uniform magnetic field of 50 mT.

- (i) Calculate the induced emf in the copper rod.
- (ii) Calculate the magnitude of induced current through the copper rod.
- (iii) State a direction of induced current through the copper rod.

[5 marks]

5. (a) An object is placed at a position of 10.0 cm in front of a mirror. Image formed in the mirror is upright and its size is twice the size of the object.
 - (i) State whether this mirror is a convex mirror or a concave mirror and determine whether the image is real or virtual.
 - (ii) Calculate the radius of curvature of the mirror.
 - (iii) Sketch a labelled ray diagram to show the formation of the image.

[8 marks]

- (b) (i) Show that the linear magnification m of a converging lens of focal length f for an image distance v is given by

$$m = -\frac{v}{f} + 1$$

- (ii) A diverging lens with focal length of 60 cm forms an upright image with a linear magnification of $\frac{1}{3}$. Calculate the image distance.

[6 marks]

6. (a) In Young's interference experiment two slits are illuminated with orange light of wavelength $6.0 \times 10^{-7}\ \text{m}$. The interference pattern is observed on a screen which is located 3.0 m from the slits. The eighth dark fringe pattern is observed 7.25 mm from the central bright.

- (i) Calculate the path difference for the light at the fourth bright fringe.
- (ii) Calculate the distance between the fifth order maximum fringe from the central bright.
- (iii) Two lights with wavelength λ and 600 nm incidents on the same double slit and same distance between slits and screen. The second dark fringe of the wavelength 600 nm light is formed at the same position on the screen as the third bright fringe of the wavelength λ . Determine λ .

[8 marks]

- (b) In a Young's double-slits experiment, the slits are 0.3 mm apart and the distance of the screen to the slits is 50 cm. The slits are illuminated by an orange light of wavelength 6.0×10^{-7} m.
- (i) Calculate the separation between the adjacent bright fringes.
 - (ii) If the orange light is replaced with a violet light ($\lambda = 420$ nm), state the effect of the fringe separation. Explain your answer.
 - (iii) If the distance between the screen and the slits is increased, describe the observation on the screen. Explain your answer.

[6 marks]

END OF QUESTION PAPER